

Alkali Metals

As well as hydrogen, the first column of the periodic table contains the alkali metals, a group of six elements that react vigorously with water to produce chemicals known as alkalis. Members of this group are soft enough to cut with a knife. They are never found pure in nature, but always as a compound, from which the pure metal is extracted.

Blue-violet flame of burning caesium

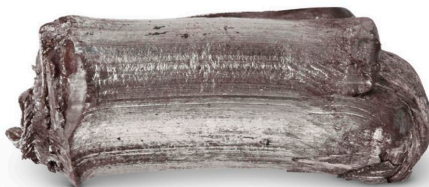
Coloured flames

Each alkali metal burns with a distinctive colour. While lithium burns with deep-red flames, potassium's flames are lilac and sodium's are yellow. Chemists look for these colours to identify the metals involved in reactions.

Piece of burning caesium

Easily tarnished

Freshly cut or scraped alkali metals are shiny, but turn dull as their surface reacts with oxygen in the air and forms a thin layer of the metal's oxide. In laboratories, the metals are stored in oil to prevent this.



Oxide layer gives this piece of lithium a dull appearance.

Highly reactive

Alkali metals are usually stored in oil because they are so reactive to the oxygen and water vapour in the air that some of them will burst into flames when they come into contact with it. Lithium tarnishes in the air, but doesn't ignite until it is heated (as shown on the left). It combines with oxygen to form a whitish compound called lithium oxide.

